THE DESIGN PROCESS

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The design process in HCI is a user-centered, iterative approach that ensures interactive systems are usable, accessible, efficient, and satisfying. Unlike traditional software design that focuses on functionality alone, HCI design emphasizes how users think, act, and feel when interacting with technology.

INTERACTION DESIGN BASICS

Interaction Design is the discipline of designing interactive digital products that support users in achieving their goals effectively and enjoyably.

It answers the questions:

- How does the user interact with the system?
- What happens when the user performs an action?
- How does the system respond?

CORE COMPONENTS OF INTERACTION DESIGN

1. Users – their abilities, limitations, preferences, and experience levels
2. Tasks – what users want to accomplish
3. Context of Use – environment, device, time pressure, and social setting

FIVE DIMENSIONS OF INTERACTION DESIGN

1. Words – labels, instructions, error messages
2. Visual Representations – icons, typography, layout, color
3. Physical Objects / Space – devices, screen size, input methods
4. Time – response time, animations, transitions
5. Behavior – how the system reacts to user actions

GOALS OF INTERACTION DESIGN

1. Effectiveness – tasks can be completed correctly
2. Efficiency – tasks are completed quickly
3. Safety – minimizes errors and harm
4. Utility – provides needed functions
5. Learnability & Memorability – easy to learn and remember

HCI IN THE SOFTWARE PROCESS

HCI is not a single phase—it is embedded throughout the Software Development Life Cycle (SDLC).

HCI IN THE SOFTWARE PROCESS

Implementation

- Applying usability principles during coding
- Consistent UI components
- Accessibility implementation (ARIA labels, contrast, keyboard access)

Testing

- Usability testing
- Heuristic evaluation
- User acceptance testing (UAT)

HCI IN THE SOFTWARE PROCESS

Deployment & Maintenance

- Collecting real-world user feedback
- Continuous improvement
- Redesign based on evolving user needs

DESIGN RULES

Design rules are established knowledge that help designers avoid usability problems.

- Design Principles

DESIGN PRINCIPLES

General rules derived from cognitive psychology and usability studies.

Visibility

- Users should easily see available actions

→ Example: visible buttons instead of hidden menus

Feedback

- System must inform users about the result of their actions
- → Example: “Data saved successfully”

Consistency

- Similar actions should have similar results
- → Same icon meaning across the system

DESIGN PRINCIPLES

General rules derived from cognitive psychology and usability studies.

Affordance

- Design suggests how something should be used

→ A button looks clickable

Constraints

- Limit actions to prevent errors

→ Disabled submit button until form is complete

STANDARDS

Formal and internationally accepted rules.

Examples:

- ISO 9241 – Ergonomics of human-system interaction
- WCAG – Web Content Accessibility Guidelines

GUIDELINES

Platform-specific recommendations.

Examples:

- Android Material Design
- Apple Human Interface Guidelines
- Microsoft Fluent Design

UNIVERSAL DESIGN

Universal Design in HCI means designing systems that can be used by everyone, not only the “average” user.

Why Universal Design Matters

- Users may have:
  - Visual, hearing, or motor impairments
  - Temporary disabilities (injury, fatigue)
  - Situational limitations (bright sunlight, noisy environments)

SEVEN PRINCIPLES OF UNIVERSAL DESIGN

Equitable Use

→ Same access for all users

- Example: captions for videos

Flexibility in Use

- → Multiple ways to interact
- Example: mouse, keyboard, touch

SEVEN PRINCIPLES OF UNIVERSAL DESIGN

Simple and Intuitive Use

- → Easy to understand regardless of experience
- Example: step-by-step forms

Perceptible Information

- → Information communicated effectively
- Example: text + icons + color contrast

SEVEN PRINCIPLES OF UNIVERSAL DESIGN

Tolerance for Error

- → Minimize consequences of mistakes
- Example: undo and confirmation dialogs

Low Physical Effort

- → Minimal fatigue
- Example: large touch targets

Size and Space for Use

- → Accommodates different devices and users
- Example: responsive layouts

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THANK YOU